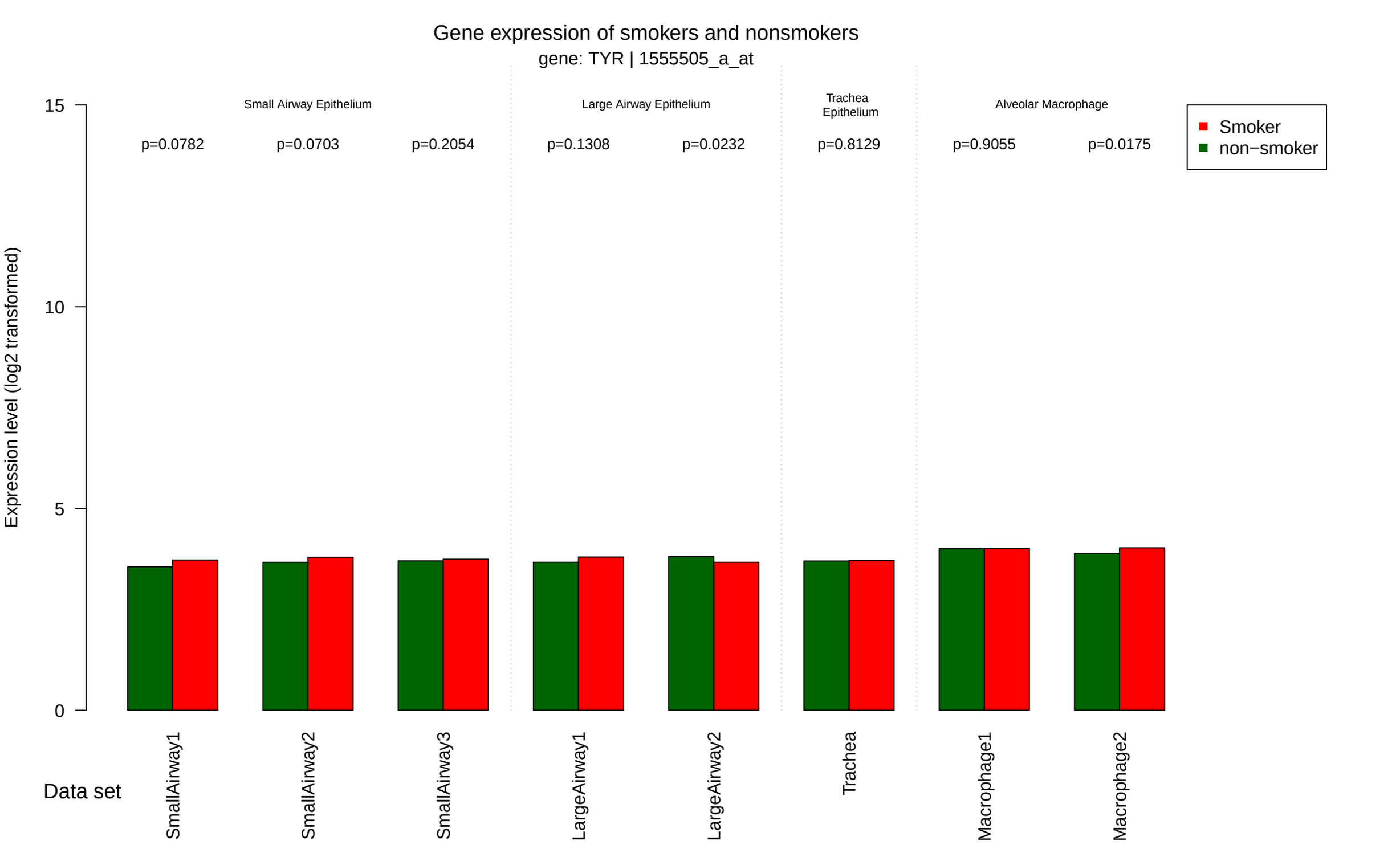
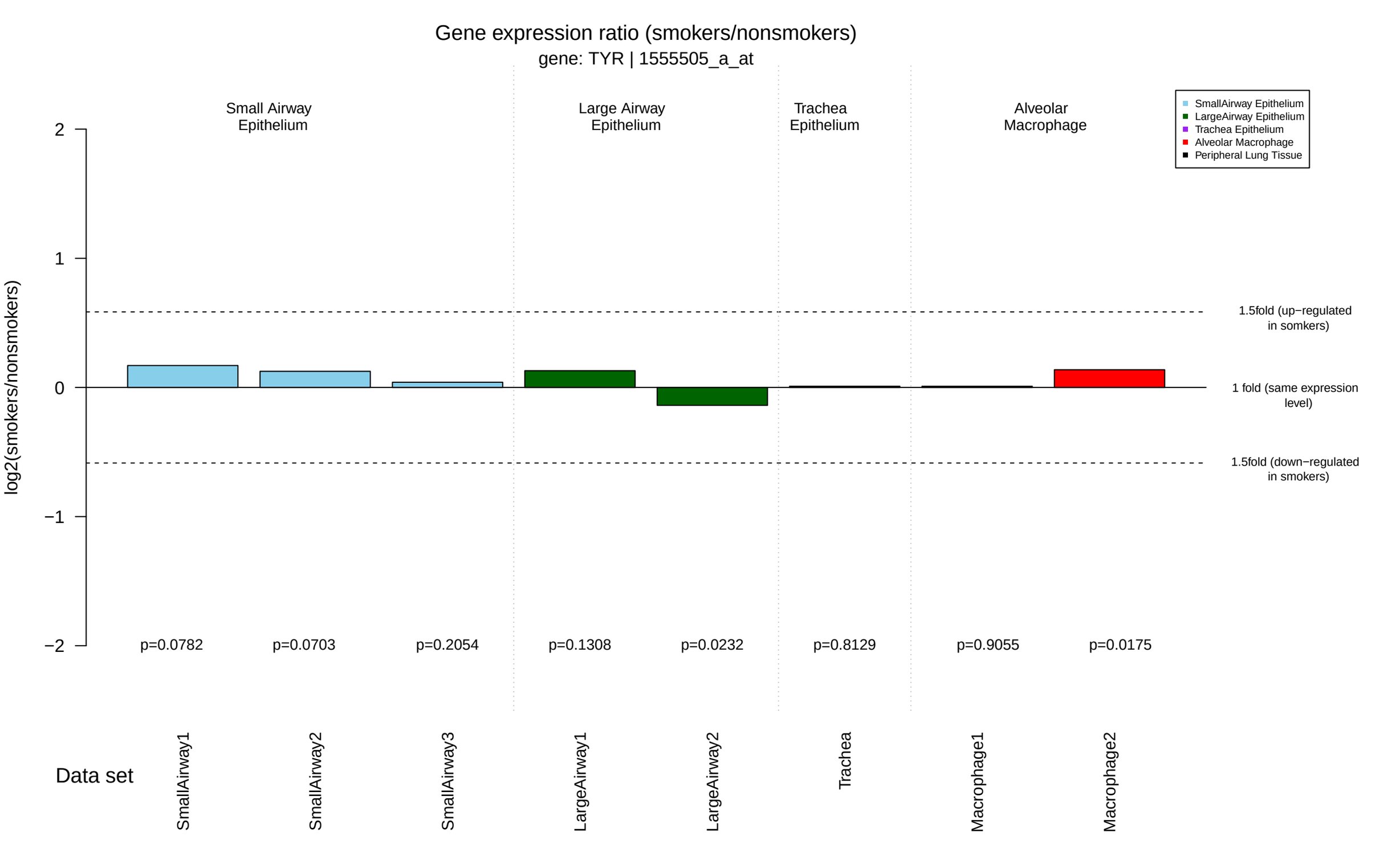
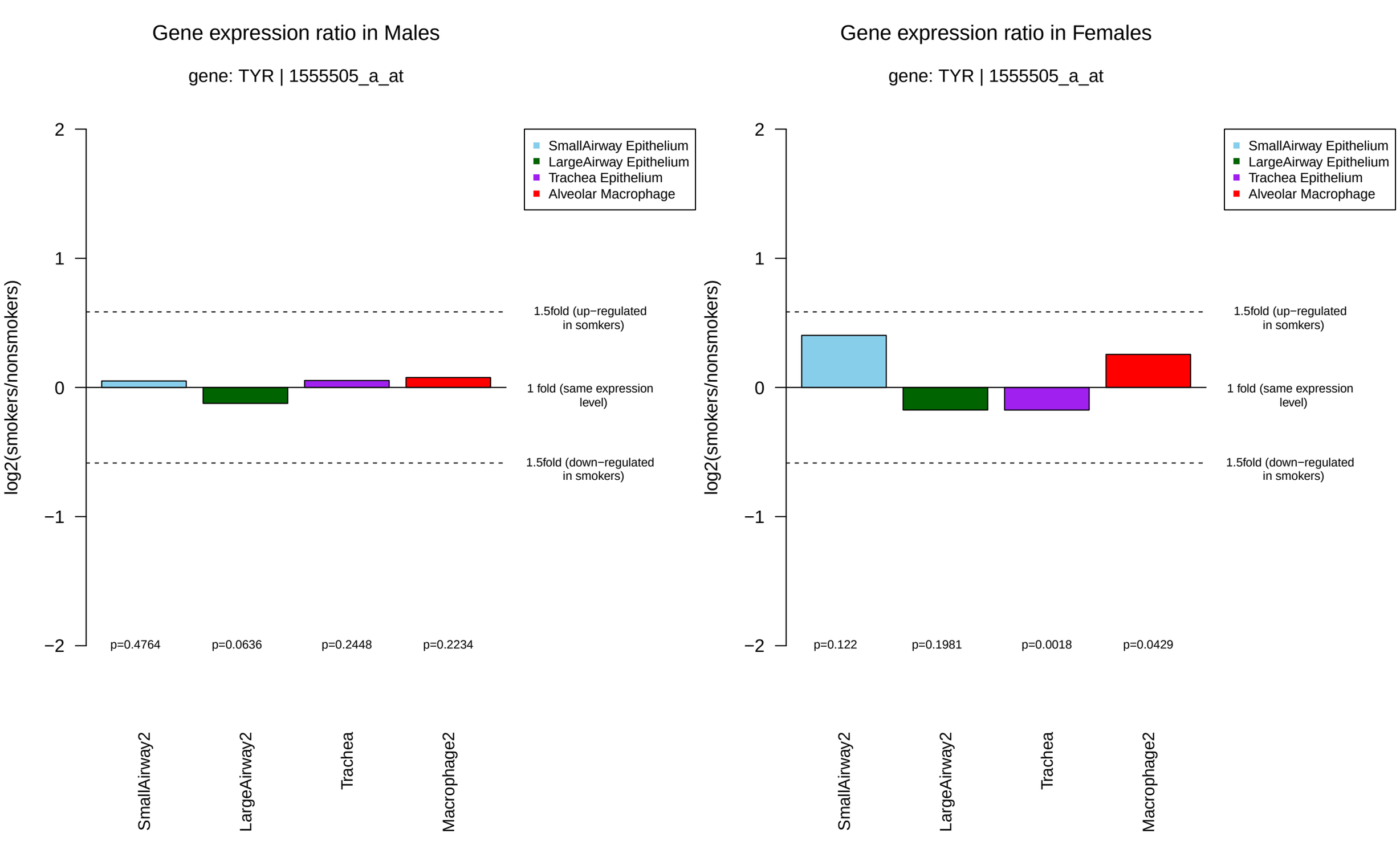
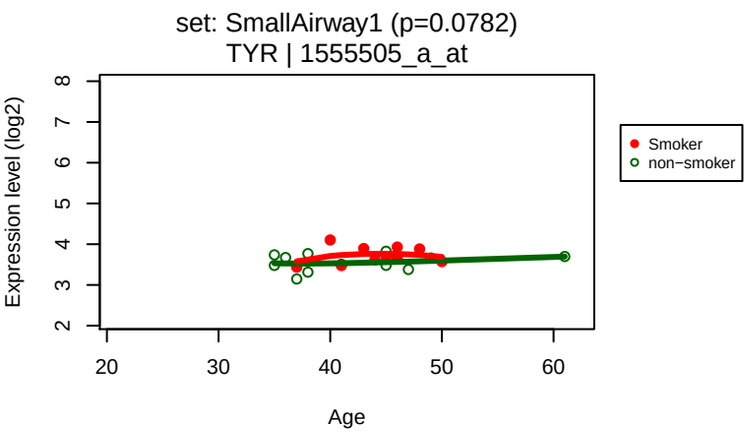
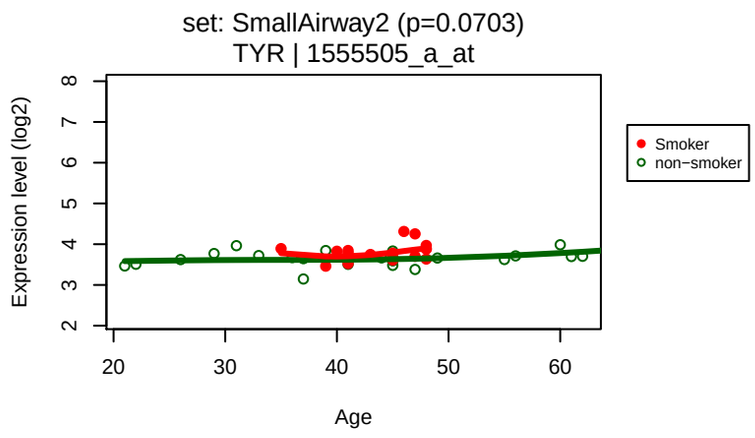
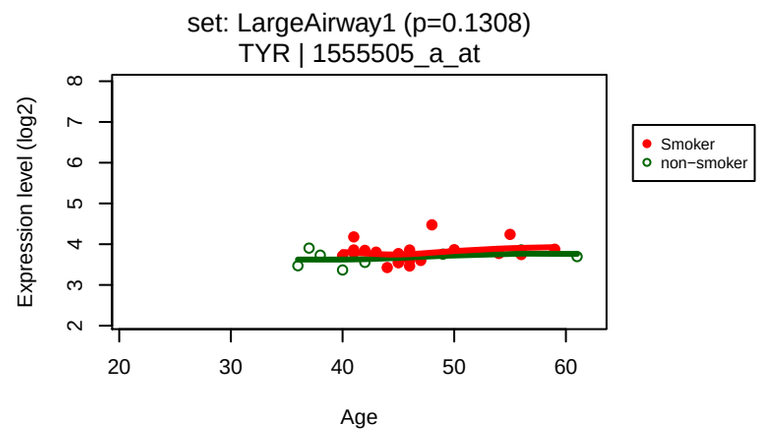

Gene expression of smokers and nonsmokers
gene: TYR | 1555505_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0782
p=0.0703
p=0.2054
p=0.1308
p=0.0232
p=0.8129
p=0.9055
p=0.0175
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

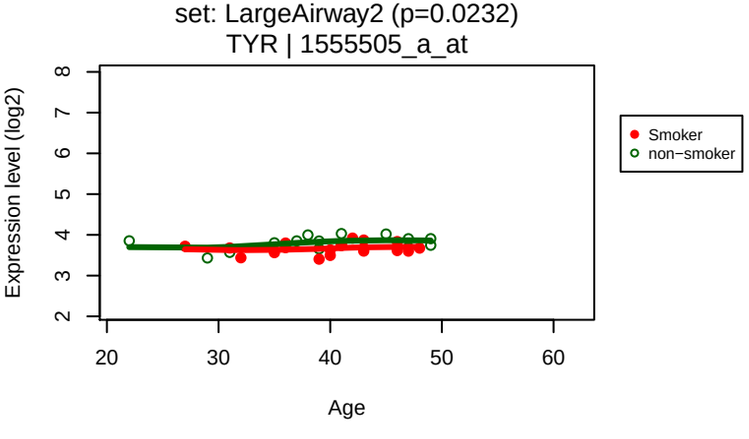

Gene expression ratio (smokers/nonsmokers)
gene: TYR | 1555505_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0782
p=0.0703
p=0.2054
p=0.1308
p=0.0232
p=0.8129
p=0.9055
p=0.0175
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

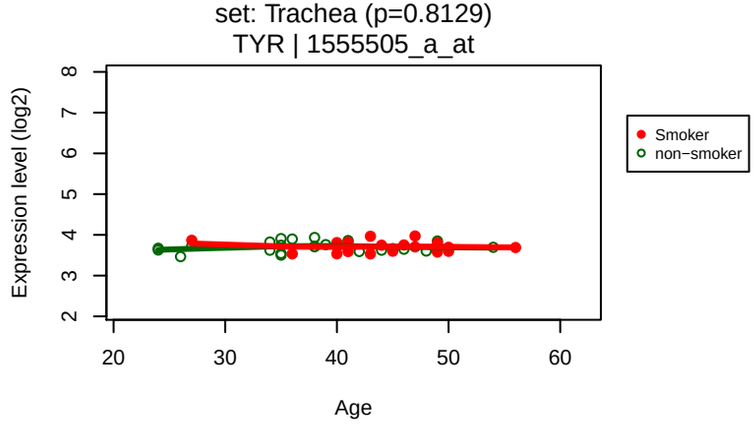

Gene expression ratio in Males
gene: TYR | 1555505_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4764
p=0.0636
p=0.2448
p=0.2234
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TYR | 1555505_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.122
p=0.1981
p=0.0018
p=0.0429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

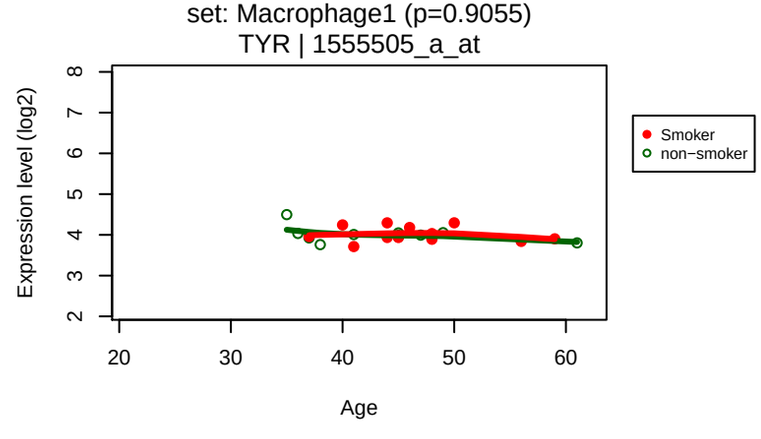

set: SmallAirway1 (p=0.0782)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker

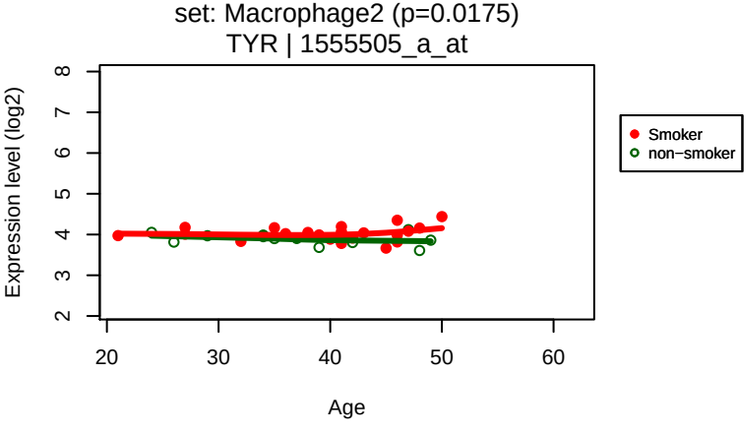

set: SmallAirway2 (p=0.0703)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker

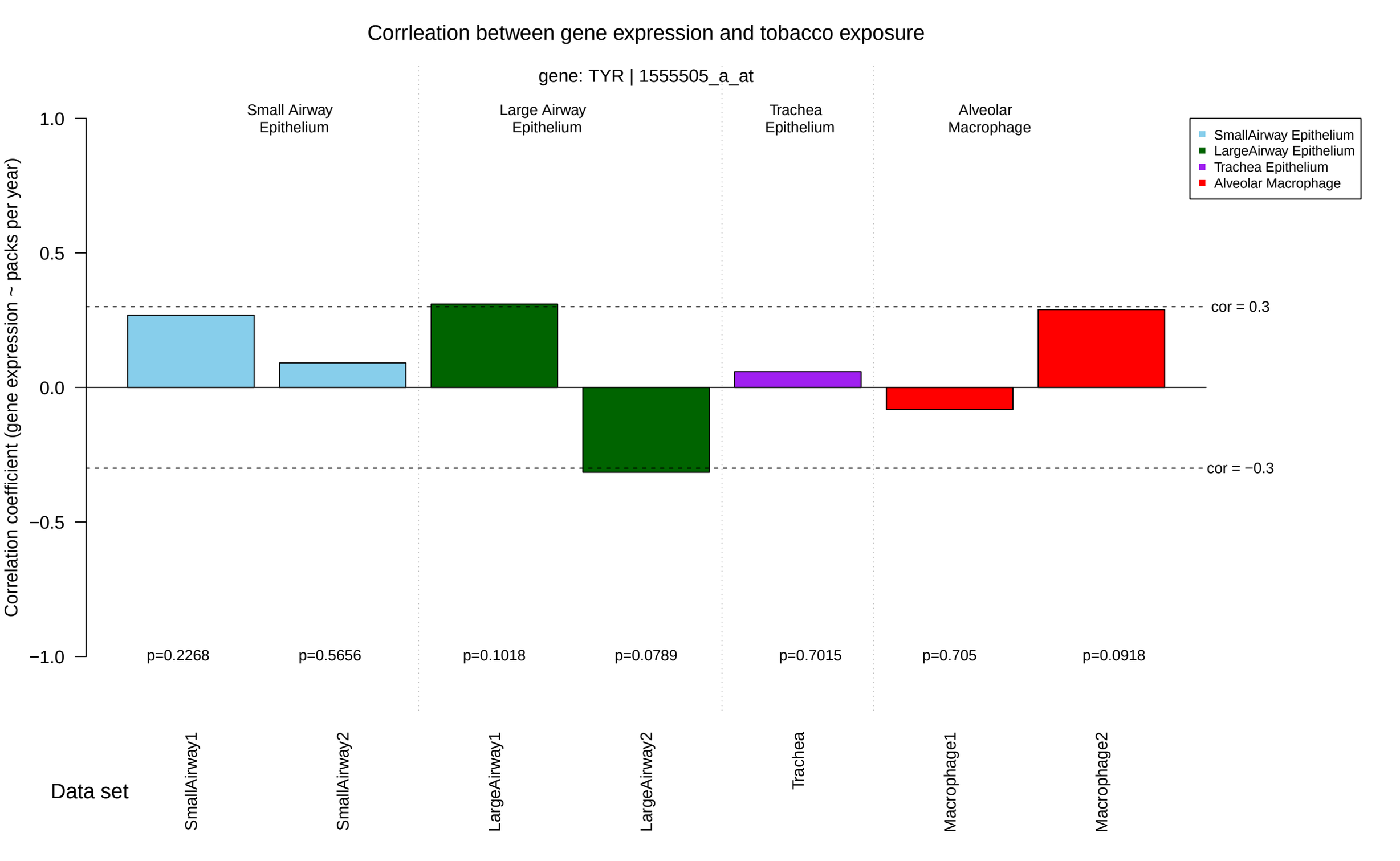

set: LargeAirway1 (p=0.1308)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0232)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8129)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9055)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0175)
TYR | 1555505_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TYR | 1555505_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2268
p=0.5656
p=0.1018
p=0.0789
p=0.7015
p=0.705
p=0.0918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2